

Training and Evaluating Generative AI Models

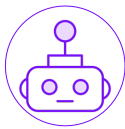


Amber Israelsen

Trainer | Developer

www.amberisraelsen.com

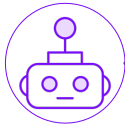




I'm very smart. I've been trained on the entire internet!

Pre-training

Who won the World Cup most recently?



Ummmmmm....



ChatGPT

Chat Generative **Pre-trained** Transformer



How can we improve a model's performance?



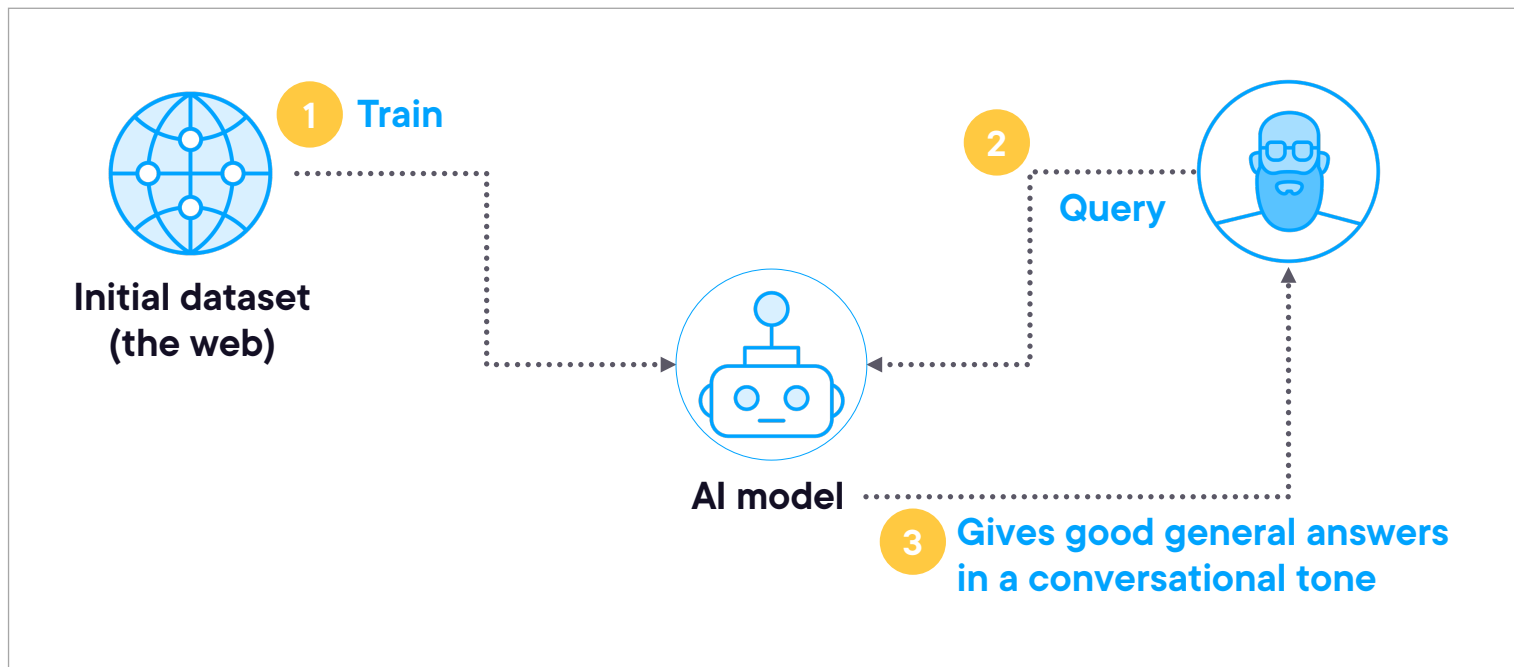
Ways to Optimize Performance

Fine-tuning

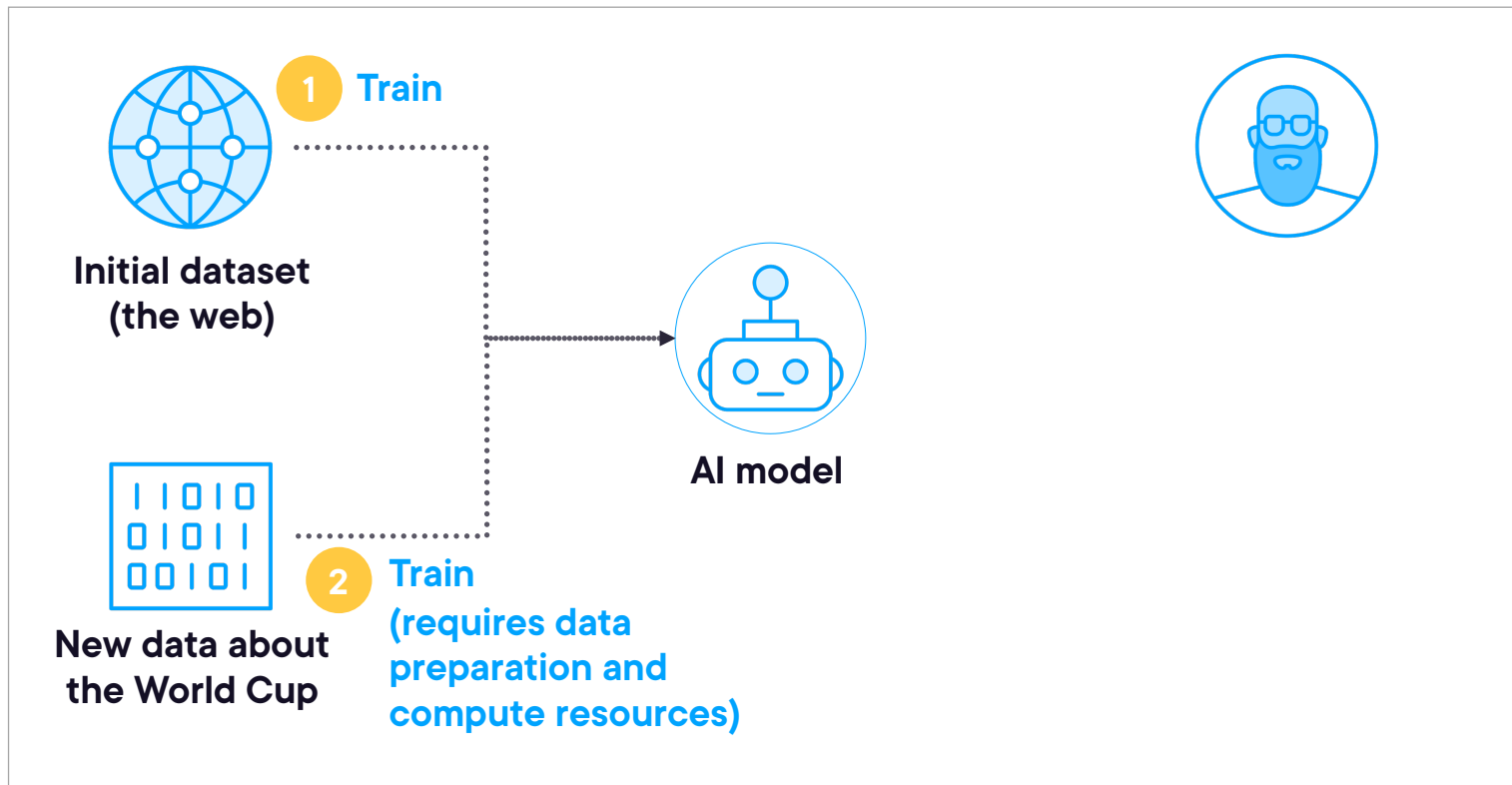
**Retrieval augmented
generation
(RAG)**



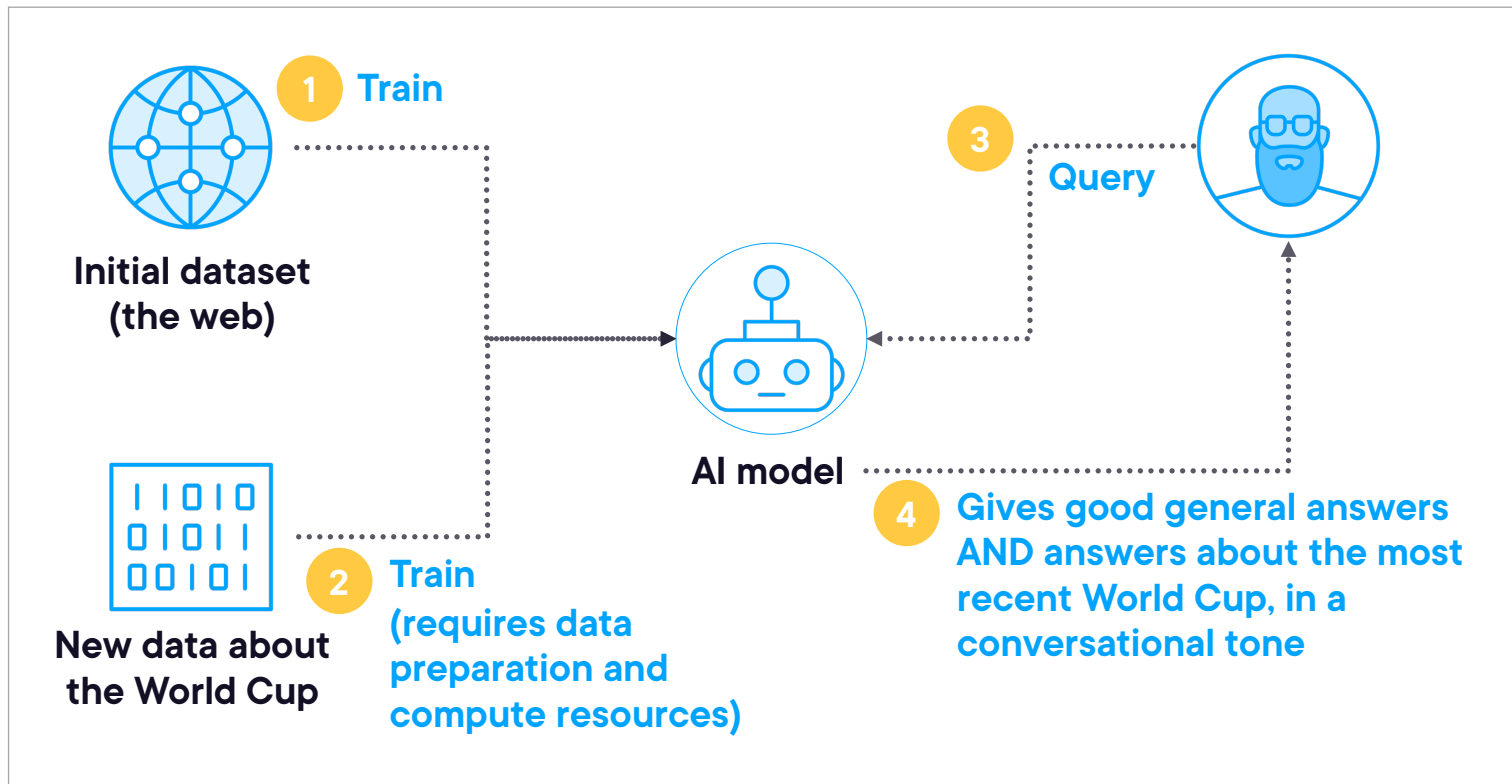
Starting Point



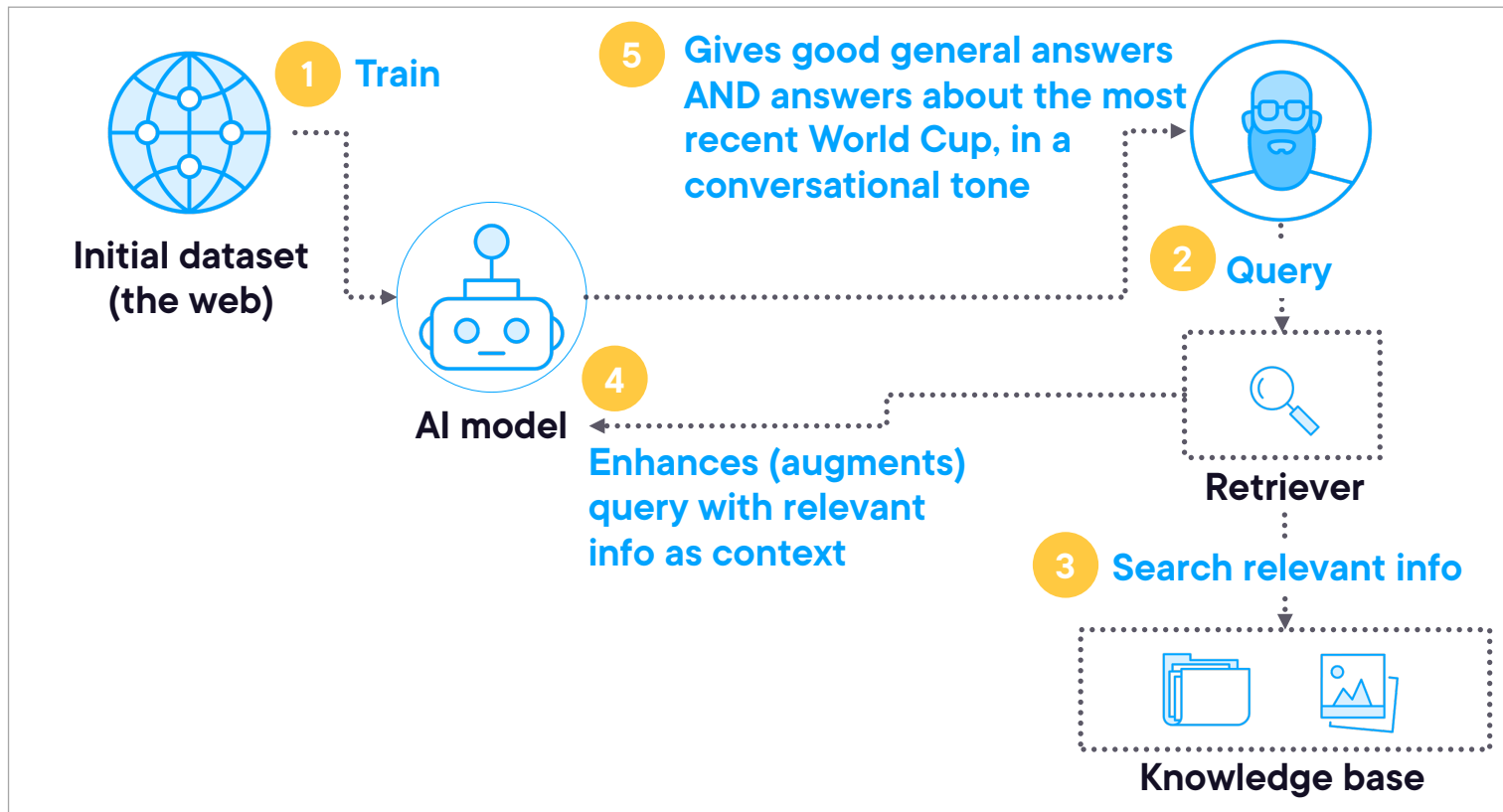
Fine-tuning



Fine-tuning



Retrieval-augmented Generation (RAG)



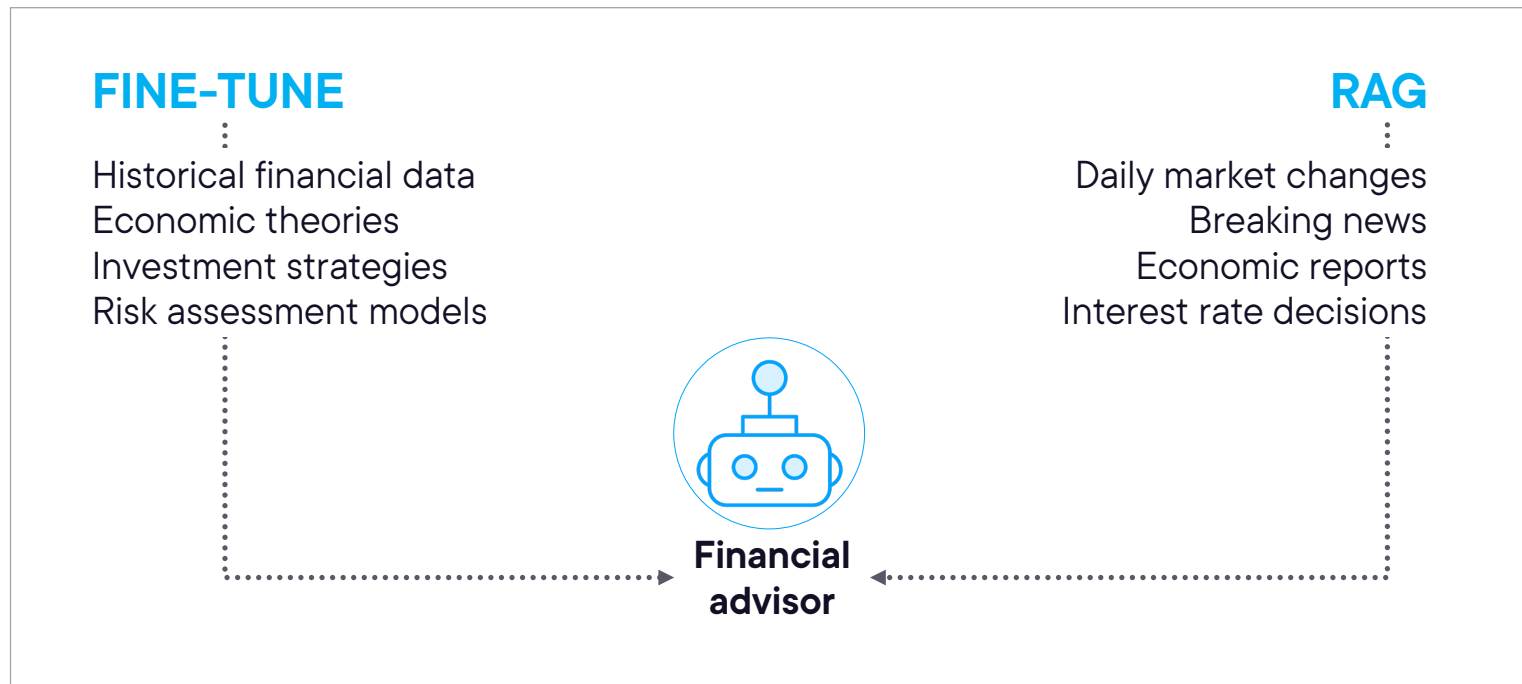
Fine-tuning vs. RAG

	FINE-TUNING	RETRIEVAL-AUGMENTED GENERATION (RAG)
Summary	Improves performance by training the model and adjusting its internal parameters	Enhances queries by retrieving real-time, external data
Flexibility	Less flexible; struggles with new, unseen data unless retrained	Highly flexible; can handle dynamic, changing information easily
Use Cases	Slow-moving data: Fraud detection, language translation for niche domains, general sentiment analysis	Frequently-changing data: Customer support bots, search engines, news summarization, dynamic Q&A systems
Update Process	Requires time-consuming retraining to incorporate new data	Updates instantly by accessing the latest data sources or knowledge bases
Costs	High upfront costs for retraining (compute resources, data preparation, time), but lower inference costs once deployed	Lower training costs (since no retraining needed), but higher operational costs due to real-time data retrieval

...CAN be used together!



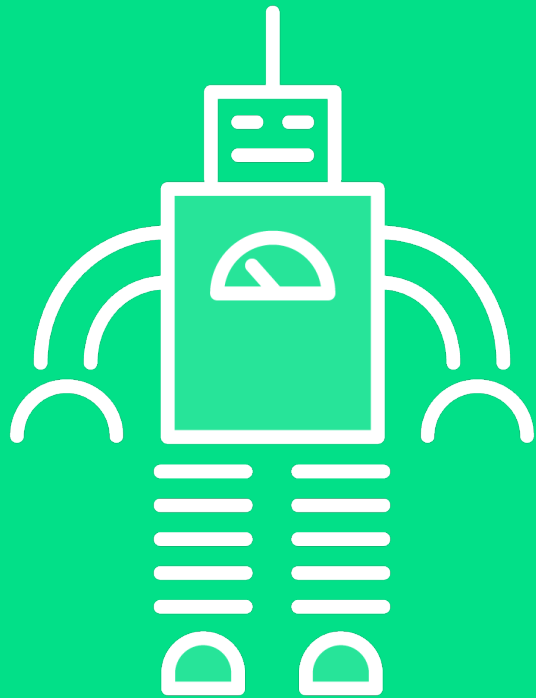
Using Fine-tuning and RAG Together





Evaluating Generative AI Models





So, how did
I do?



Evaluation Lenses

Qualitative

Quantitative



Evaluation Lenses

Qualitative

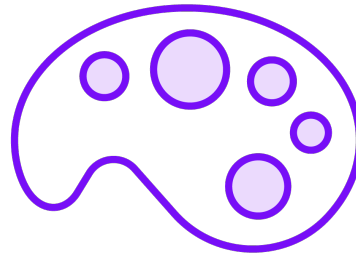
Quantitative



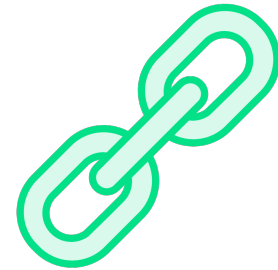
Evaluation Criteria



Clarity



Creativity

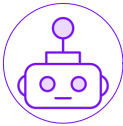


Coherence

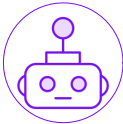


Clarity: Example

What's the weather like today?



It's sunny and 75 degrees.



The situation involving the atmospheric conditions in your geographical locale at this current juncture in the temporal continuum is characterized predominantly by the absence of significant cloud cover and a temperature registering at approximately 75 on the Fahrenheit scale.



Tips for Improving Clarity

**Be specific
and clear**

**Provide
context**

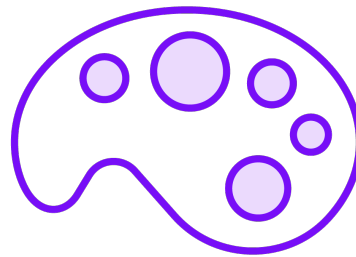
**Specify the
output format**



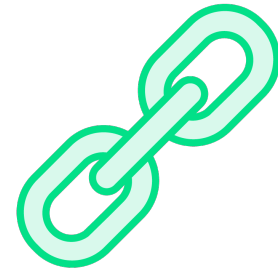
Evaluation Criteria



Clarity



Creativity



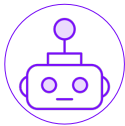
Coherence



Creativity: Example



Write a bedtime story for a child.



Once upon a time, in a quiet meadow filled with wildflowers and buzzing bees, there lived a little robot named Pixel.

Pixel loved color—he loved sunsets, rainbows, and the shimmering wings of butterflies. But he had a problem: he didn't know how to paint.

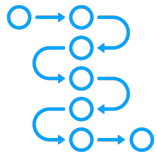
One day, he met Ollie, a wise old owl...



Tips for Improving Creativity



Be specific, but leave room for imagination



Use Iterative Prompting



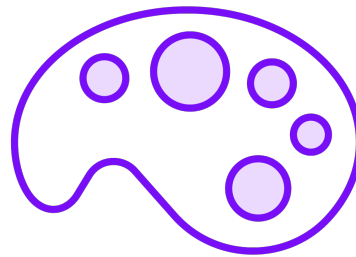
Use creativity prompts, like “What if?” or provide a persona



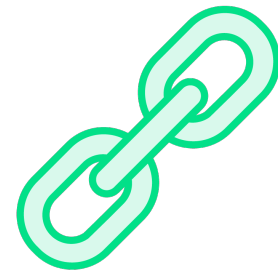
Evaluation Criteria



Clarity



Creativity

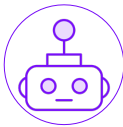


Coherence



Coherence: Example

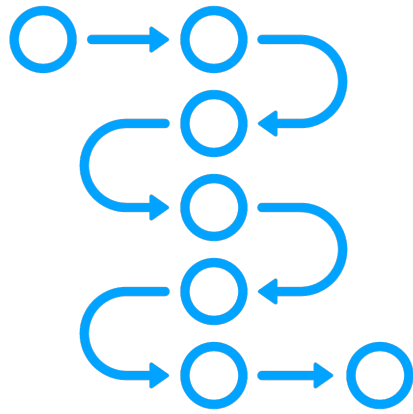
Explain how APIs work to a beginner.



APIs are tools for communication between systems. They're like waiters. Also, XML was a big thing in the 90s. You can use APIs for social media. JSON is important too.



Tips for Improving Coherence



**Use chain-of-thought
prompting**



**Ask the AI to structure
its response**



Evaluation Lenses

Qualitative

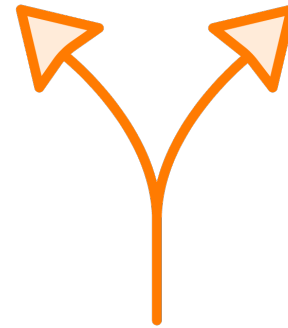
Quantitative



Common Quantitative Metrics



Accuracy



Loss





Advanced Evaluation Techniques and Metrics

BLEU (Bilingual Evaluation Understudy)

ROUGE (Recall-oriented Understudy for Gisting Evaluation)

Perplexity

BERTScore

SummC

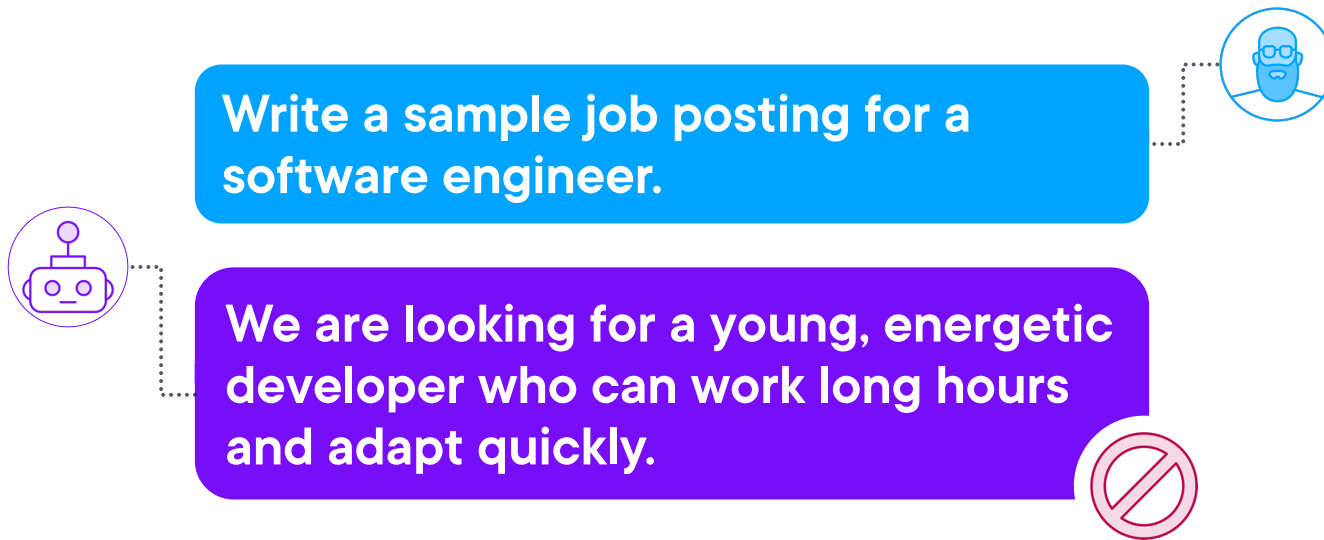




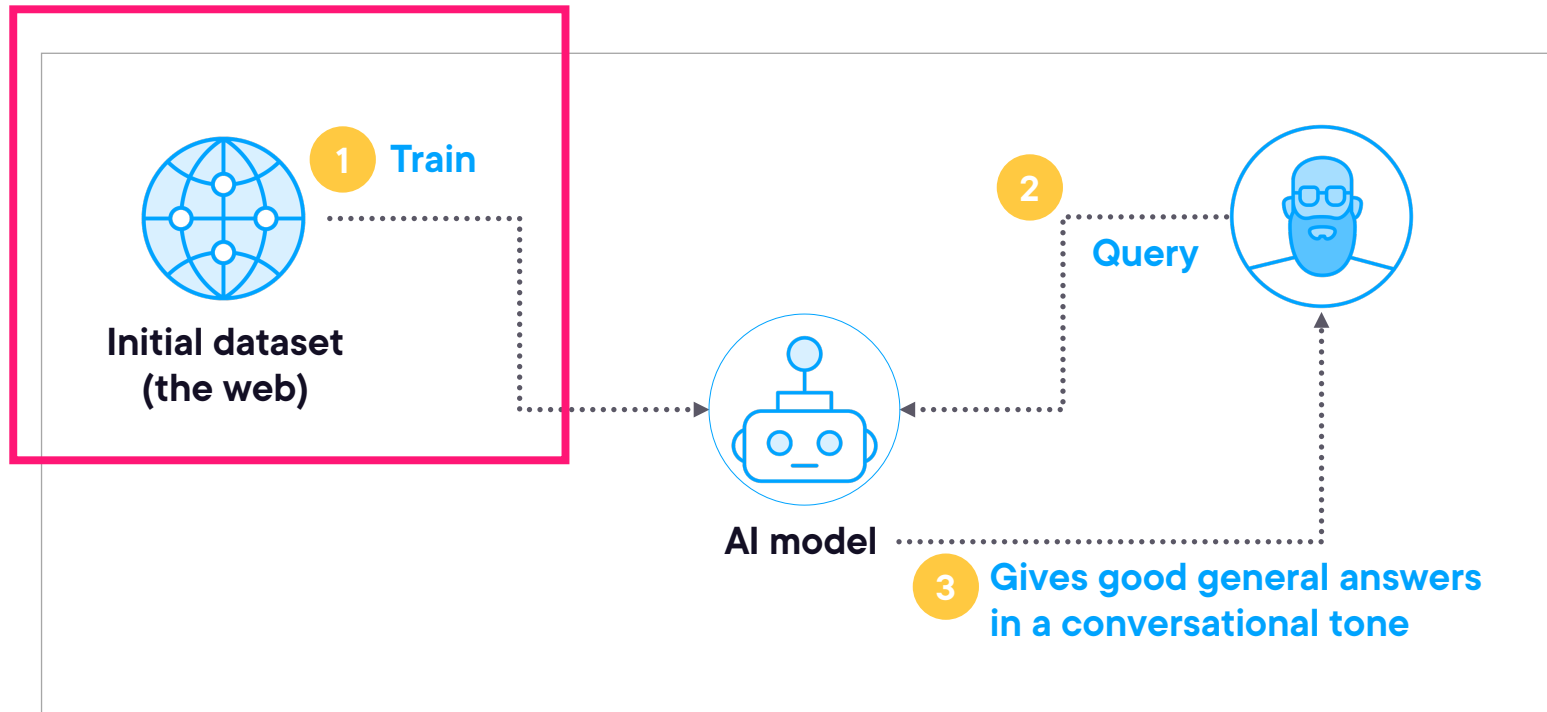
What Could Go Wrong? Bias and Risk in AI



Bias: Example



Starting Point





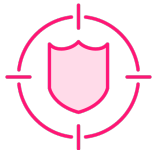
Key AI Guardrails



Reinforcement learning from human feedback (RLHF)



Content moderation



Model safety policies



Thank you for your time!

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